Network device

1. Explain behaviour hub device?
   - It is used to connect multiple network devices.

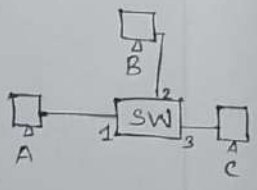
For example:

- Imagine 1 hub and 3 Pc's connected to hub. If [Data] A/B If A send data means it will go to hub first and hub will broadcast. so B also can see the data and C also can see the data but B send response to A [Data] B/A.
- Imagine Pc 1 is sending data and at the same time Pc 3 is also sending data there will be traffic and data will be loss and this problem is called collision.
- To overcome this problem switch was introduced.

2. Explain behaviour of switch device?
   - It is used to connect multiple network devices.
   - It contains mac table
   - It acts as both broadcast and unicast

For example:

If A send data means it will first do address learning that is it store port no (1) and source mac in mac table. In mac table destination mac is available means it will unicast otherwise it will broad cast.

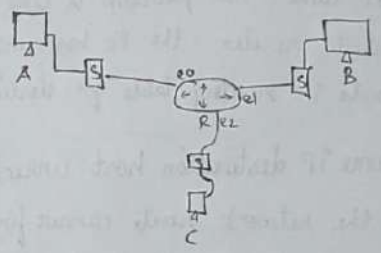


	MAC table
1	A
2	B
3	C

- In mac table data stored for "5 min"
- If A send data means it will broadcast called single broadcast domain.
- To overcome this problem "router" is introduced

3. Explain behaviour of Router device?
   - It is used to break broadcast domain
   - It is only unicast
   - It contains routing table
   - It is used to connect different network.

For example:



- If A send data to its destination. It will first send to switch. The switch will broadcast and then it send to router. In the routing table the directly connected router is there means it will unicast.
- Routing table

212.118.116.0\24 - e0	In routing table N/w ID, Prefix length, Interface number is stored.
179.202.0.0\16 - e1	
9.0.0.0\8 - e2	

- We have assign IP for router also defaultly we have set IP like 9.0.0.1. It is not compulsory. for our forget purpose.